

Digital Parking Permit System - SER216 Final Project

GitHub Repository Link: https://github.com/elzodxon/SER216_Final_Project_Elzodxon

Task 1: Requirements

Functional Requirements:

1. The system shall allow students and staff to register vehicles using plate number, owner ID, and vehicle type.
2. The system shall allow users to request parking permits for available parking zones.
3. The system shall display available parking zones before permit approval.
4. The system shall allow security staff to verify permits and update violation records.

Non-Functional Requirements:

1. The system shall verify parking permits within 3 seconds during security checks.
2. The system shall protect user and vehicle data using secure authentication and access control.

Task 2: Workflow Diagram

Vehicle Registration -> Permit Request -> Zone Check -> Permit Approval -> Parking Verification

Diagram file: [diagrams/workflow_diagram.png](#)

Task 3: Test Case Table

TC01 System Test: Complete permit process with valid student, vehicle, and zone. Expected: approved permit.

TC02 System Test: Security verifies an approved valid permit. Expected: permit is valid.

TC03 Acceptance Test: Student requests permit for available zone. Expected: approved by university rules.

TC04 Acceptance Test: User checks parking zones. Expected: correct available zones shown.

TC05 Regression Test: Valid non-expired permit. Expected: not shown as expired.

TC06 Regression Test: Security adds violation. Expected: saved without affecting permit status.

Task 4: Defect Analysis

Defect: A valid parking permit is shown as expired during security verification.

Defect category: Functional defect.

Severity: High.

Priority: High.

Possible root cause: Incorrect date comparison, timezone mismatch, database expiry mismatch, or cached permit status.

Regression test: Verify that a valid permit with a future expiry date is displayed as valid during security verification.

Task 5: GitHub Evidence Checklist

Screenshots required: repository homepage, README.md, commit history with at least 3 commits, feature branch, pull request.

Screenshot files are stored in the screenshots folder. Replace placeholder files with real GitHub screenshots if your instructor requests.